

Skills Summary

Graphs, Line Plots, and Shape Sorting

Use this reference while completing your worksheet or when studying.

Every question on the worksheet starts the same way: decide what the display is showing you, then decide whether the question wants you to *read* one number, *add* several, *compare* two, or *sort* shapes first and count afterwards.

1. Reading a bar graph

How to read it: each bar stands for one group. Slide from the top of the bar across to the number scale — that number is the count for that group.

Read one bar for “how many did ___ have?”. **Add every bar** for “altogether” or “in all”. **Subtract two bars** for “how many more” or “how many fewer”.

Example: A bar graph shows fruit picked: apples 7, pears 4, plums 9. How many pieces of fruit were picked in all?

Step 1. “In all” means every group counts, so this is an adding question, not a reading-one-bar question.

Step 2. $7 + 4 + 9$: first $7 + 4 = 11$, then $11 + 9 = 20$.

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Try it: A bar graph shows stickers: stars 6, moons 3, suns 8. How many stickers in all?

check: 17

2. Reading a line plot

How to read it: the numbers along the bottom are *measurements*. Each X above a number stands for *one object* of that size.

So a stack of 5 X marks above the 4 means five objects, each 4 units long. To find how many objects there are altogether, count every X on the whole plot.

Example: A line plot of ribbon lengths has 3 X marks above 2, 4 X marks above 3, and 2 X marks above 4. How many ribbons are 3 inches long?

Step 1. The question names one length, so find that number on the line and count only the stack sitting above it.

Step 2. Above the 3 there are 4 X marks, and each X is one ribbon.

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Try it: A line plot of book widths has 2 X marks above 5, 6 X marks above 6, and 3 X marks above 7. How many books are 6 cm wide?

check: 9

3. Comparing and measuring a data set

Most common (mode): the measurement with the *tallest* stack of X marks. In the list 2, 3, 3, 3, 5 the most common value is 3.

Range: greatest measurement – least measurement. It tells you how spread out the data is.

(!) Both answers are measurements, not stack heights. Give the number from the bottom of the plot, with its unit.

Example: Ribbon lengths in inches: 4, 4, 6, 7, 7, 9. Find the range.

Step 1. Range compares the two ends of the data, so I only need the greatest and the least value — the repeats in the middle do not matter.

Step 2. Greatest is 9, least is 4, and $9 - 4 = 5$.

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Try it: Pencil lengths in centimetres: 2, 5, 5, 6, 8. Find the range.

9 :pəp

4. Sorting shapes by sides and corners

Count the sides; the corners always match. 3 sides = triangle, 4 sides = quadrilateral, 5 sides = pentagon, 6 sides = hexagon.

Squares, rectangles, rhombuses and trapezoids are all quadrilaterals. A square is the special one that has 4 *equal* sides *and* 4 square corners. So “it has 4 sides” never proves “it is a square”.

Example: A bag holds a triangle, a square, a rectangle, a trapezoid and a hexagon. How many of them have exactly 4 sides?

Step 1. Sorting comes before counting: give every shape its side number first, then count only the row I was asked about.

Step 2. Triangle 3, square 4, rectangle 4, trapezoid 4, hexagon 6 — so the “4 sides” row holds the square, the rectangle and the trapezoid.

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Try it: A bag holds a rectangle, a rhombus, a pentagon and a triangle. How many have exactly 4 sides?

2 :pəp